

- Renally excreted
- Active metabolite (carbamazepine-10,11 epoxide)
- Initial half-life 26–65 hours (35–40 hours for extended-release formulation); half-life 12–17 hours with repeated doses
- Half-life of active metabolite is approximately 34 hours
- * Is not only a substrate for CYP450 3A4, but also an inducer of CYP450 3A4
- * Thus, carbamazepine induces its own metabolism, often requiring an upward dosage adjustment
- Is also an inducer of CYP450 2C9 and weakly of 1A2 and 2C19
- Food does not affect absorption



Drug Interactions

- Enzyme-inducing antiepileptic drugs (carbamazepine itself as well as phenobarbital, phenytoin, and primidone) may increase the clearance of carbamazepine and *lower* its plasma levels
- CYP450 3A4 inducers, such as carbamazepine itself, can *lower* the plasma levels of carbamazepine
- CYP450 3A4 inhibitors, such as nefazodone, fluvoxamine, and fluoxetine, can *increase* plasma levels of carbamazepine
- Carbamazepine can *increase* plasma levels of clomipramine, phenytoin, primidone

- Carbamazepine can *decrease* plasma levels of acetaminophen, clozapine, benzodiazepines, dicumarol, doxycycline, theophylline, warfarin, rivaroxaban, apixaban, dabigatran, edoxaban, and haloperidol as well as other anticonvulsants such as phenoximide, methsuximide, ethosuximide, phenytoin, tiagabine, topiramate, lamotrigine, and valproate
- Carbamazepine can *decrease* plasma levels of hormonal contraceptives and adversely affect their efficacy
- Combined use of carbamazepine with other anticonvulsants may lead to altered thyroid function
- Combined use of carbamazepine and lithium may increase risk of neurotoxic effects
- Depressive effects are increased by other CNS depressants (alcohol, MAOIs, other anticonvulsants, etc.)
- Combined use of carbamazepine suspension with liquid formulations of chlorpromazine has been shown to result in excretion of an orange rubbery precipitate; because of this, combined use of carbamazepine suspension with any liquid medicine is not recommended

Other Warnings/Precautions

* Patients should be monitored carefully for signs of unusual bleeding or bruising, mouth sores, infections, fever, or sore throat, as the risk of aplastic anemia and agranulocytosis with carbamazepine use is 5–8 times greater than in the general population (risk in the

untreated general population is 6 patients per 1 million per year for agranulocytosis and 2 patients per 1 million per year for aplastic anemia)

- Because carbamazepine has a tricyclic chemical structure, use with caution with MAOIs, including 14 days after MAOIs are stopped (for the expert)
- May exacerbate angle-closure glaucoma
- Because carbamazepine can lower plasma levels of hormonal contraceptives, it may also reduce their effectiveness
- May need to restrict fluid intake because of risk of developing syndrome of inappropriate antidiuretic hormone secretion, hyponatremia and its complications
- Use with caution in patients with mixed seizure disorders that include atypical absence seizures because carbamazepine has been associated with increased frequency of generalized convulsions in such patients
- Individuals with the HLA-B*1502 allele are at increased risk of developing Stevens-Johnson syndrome and toxic epidermal necrolysis
- Whenever possible, warn patients and their caregivers about the possibility of activating side effects, and advise them to report such side effects immediately

Do Not Use

- If patient has history of bone marrow suppression
- If patient tests positive for the HLA-B*1502 allele
- If there is a proven allergy to any tricyclic compound
- If there is a proven allergy to carbamazepine
- Suspension: in patients with hereditary problems with fructose intolerance

Special Populations

Renal Impairment

- Carbamazepine is renally secreted, so the dose may need to be lowered

Hepatic Impairment

- Drug should be used with caution
- Rare cases of hepatic failure have occurred

Cardiac Impairment

- Drug should be used with caution

Elderly

- Some patients may tolerate lower doses better
- Elderly patients may be more susceptible to adverse effects

Children and Adolescents

- Approved use for epilepsy; therapeutic range of total carbamazepine in plasma is considered the same for children and adults
- Ages 6–12: initial dose 100 mg twice daily (tablets) or 0.5 teaspoon (50 mg) 4 times a day (suspension); each week increase by up to 100 mg/day in divided doses (2 doses for extended-release formulation, 3–4 doses for all other formulations); maximum dose generally 1000 mg/day; maintenance dose generally 400–800 mg/day
- Ages 5 and younger: initial 10–20 mg/kg per day in divided doses (2–3 doses for tablet formulations, 4 doses for suspension); increase weekly as needed; maximum dose generally 35 mg/kg/day

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- * Use during first trimester may raise risk of neural tube defects (e.g., spina bifida) or other congenital anomalies
- Use in women of childbearing potential requires weighing potential benefits to the mother against the risks to the fetus

- * If drug is continued, perform tests to detect birth defects
- * If drug is continued, start on folate 1 mg/day early in pregnancy to reduce risk of neural tube defects
- Antiepileptic Drug Pregnancy Registry: 1-888-233-2334, www.aedpregnancyregistry.org
- Use of anticonvulsants in combination may cause a higher prevalence of teratogenic effects than anticonvulsant monotherapy
- Taper drug if discontinuing
- Seizures, even mild seizures, may cause harm to the embryo/fetus
- * For bipolar patients, carbamazepine should generally be discontinued before anticipated pregnancies
- Recurrent bipolar illness during pregnancy can be quite disruptive
- For bipolar patients, given the risk of relapse in the postpartum period, some form of mood stabilizer treatment may need to be restarted immediately after delivery if patient is unmedicated during pregnancy
- * Atypical antipsychotics may be preferable to lithium or anticonvulsants such as carbamazepine if treatment of bipolar disorder is required during pregnancy
- Bipolar symptoms may recur or worsen during pregnancy and some form of treatment may be necessary

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed
- If drug is continued while breast feeding, infant should be monitored for possible adverse effects, including hematological effects
- If infant shows signs of irritability or sedation, drug may need to be discontinued
- Some cases of neonatal seizures, respiratory depression, vomiting, and diarrhea have been reported in infants whose mothers received carbamazepine during pregnancy
- * Bipolar disorder may recur during the postpartum period, particularly if there is a history of prior postpartum episodes of either depression or psychosis
- Relapse rates may be lower in women who receive prophylactic treatment for postpartum episodes of bipolar disorder
- Atypical antipsychotics and anticonvulsants such as valproate may be safer than carbamazepine during the postpartum period when breast feeding

The Art of Psychopharmacology

Potential Advantages

- Treatment-resistant bipolar and psychotic disorders

Potential Disadvantages

- Patients who do not wish to or cannot comply with blood testing and close monitoring
- Patients who cannot tolerate sedation
- Pregnant patients

Primary Target Symptoms

- Incidence of seizures
- Unstable mood, especially mania
- Pain

Pearls

- Carbamazepine was the first anticonvulsant widely used for the treatment of bipolar disorder and is now formally approved for acute mania and mixed mania
- * An extended-release formulation has better evidence of efficacy and improved tolerability in bipolar disorder than does immediate-release carbamazepine
- Dosage frequency as well as sedation, diplopia, confusion, and ataxia may be reduced with extended-release carbamazepine
- Risk of serious side effects is greatest in the first few months of treatment
- Common side effects such as sedation often abate after a few months

- * May be effective in patients who fail to respond to lithium or other mood stabilizers
- May be effective for the depressed phase of bipolar disorder and for maintenance in bipolar disorder
- Can be complicated to use with concomitant medications

Suggested Reading

Leucht S , McGrath J , White P , Kissling W . Carbamazepine for schizophrenia and schizoaffective psychoses . *Cochrane Database Syst Rev* 2002 ;(3):CD001258 .

Marson AG , Williamson PR , Hutton JL , Clough HE , Chadwick DW . Carbamazepine versus valproate monotherapy for epilepsy . *Cochrane Database Syst Rev* 2000 ;(3):CD001030 .

Smith LA , Cornelius V , Warnock A , Tacchi MJ , Taylor D . Pharmacological interventions for acute bipolar mania: a systematic review of randomized placebo-controlled trials . *Bipolar Disord* 2007 ;**9** (6):551–60 .

Weisler RH , Kalali AH , Ketter TA . A multicenter, randomized, double-blind, placebo-controlled trial of extended-release carbamazepine capsules as monotherapy for bipolar disorder patients with manic or mixed episodes . *J Clin Psychiatry* 2004 ;**65** :478–84 .

Cariprazine

Therapeutics

Brands

- Vraylar

see index for additional brand names

Generic?

No

Class

- Neuroscience-based Nomenclature: dopamine, serotonin, norepinephrine receptor partial agonist
- Dopamine partial agonist (dopamine-serotonin partial agonist, dopamine stabilizer, atypical antipsychotic, third- generation antipsychotic; sometimes included as a second-generation antipsychotic; also a potential mood stabilizer)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia, acute and maintenance**
- **Acute mania/mixed mania**

- **Bipolar depression (bipolar I disorder)**
- Bipolar depressive episodes with mixed features (subsyndromal manic symptoms)
- Bipolar maintenance
- Major depressive episodes (unipolar) with mixed features (subsyndromal manic symptoms)
- Manic episodes with mixed features (subsyndromal depressive symptoms)
- Other psychotic disorders
- Negative symptoms of schizophrenia
- Treatment-resistant depression
- Augmentation of other antidepressants in (unipolar) major depressive disorder
- Behavioral disturbances in dementia
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- * Partial agonism at dopamine 2 receptors
- Theoretically reduces dopamine output when dopamine concentrations are high, thus improving positive symptoms and

mediating antipsychotic actions

- Theoretically increases dopamine output when dopamine concentrations are low, thus improving cognitive, negative, and mood symptoms
- Preferentially binds to dopamine 3 over dopamine 2 receptors at low doses; the clinical significance is unknown but could theoretically contribute to cariprazine's efficacy for negative symptoms. D3 partial agonism could theoretically be useful for treating cognition, mood, emotions, and reward/substance use
- Interactions at a myriad of other neurotransmitter receptors may contribute to cariprazine's efficacy
- Cariprazine also has high affinity for the serotonin 1A (partial agonist) and 2B (antagonist) receptors
- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms

How Long Until It Works

- Psychotic, and manic and depressive symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition
- Classically recommended to wait at least 4–6 weeks to determine full antipsychotic and antidepressant efficacy of drug, but in practice

some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms but does not eliminate them
- May reduce and eliminate acute manic symptoms and depressive symptoms
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis
- For second and subsequent episodes of psychosis, treatment may need to be indefinite

- Even for first episodes of psychosis, it may be preferable to continue treatment

If It Works –for Bipolar Depression

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission) or significantly reduced
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective for schizophrenia, consider higher doses or augmentation with valproate or lamotrigine

- Consider lithium and anticonvulsant mood stabilizers for mania
- Consider noncompliance and switch to another antipsychotic with fewer side effects or to an antipsychotic that can be given by depot injection
- Consider initiating rehabilitation and psychotherapy
- Consider presence of concomitant drug abuse

If It Doesn't Work –for Bipolar Depression

- Some patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)



Best Augmenting Combos for Partial Response or Treatment Resistance

- Valproic acid (valproate, divalproex, divalproex ER)
- Lamotrigine
- Topiramate
- Lithium
- Benzodiazepines

Tests

Before starting any atypical antipsychotic

- * Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- * Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)
 - has diabetes (fasting plasma glucose >126 mg/dL)
 - has hypertension (BP >140/90 mmHg)
 - has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting any atypical antipsychotic

- * BMI monthly for 3 months, then quarterly

* Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics

* Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight

- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic

* Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness and clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and cariprazine should be discontinued at the first sign of decline in WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Partial agonist actions at dopamine 2 receptors in the striatum can cause akathisia and drug-induced parkinsonism
- Partial agonist actions at dopamine 2 receptors can also cause nausea, occasional vomiting, and activating side effects
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Akathisia, drug-induced parkinsonism, restlessness
- Gastrointestinal distress
- Sedation
- Tardive dyskinesia (TD) (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- Side effects may theoretically appear several weeks after initiating cariprazine, because plasma levels and major active metabolites accumulate over time



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis
- As a class, antidepressants have been reported to increase the risk of suicidal thoughts and behaviors in children and young adults

Weight Gain



- Reported but not expected



Sedation

- Occurs in significant minority

What to Do About Side Effects

- Wait
- Wait

- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Schizophrenia: 1.5–6 mg once daily

- Bipolar mania: 3–6 mg once daily
- Bipolar depression: 1.5–3 mg once daily

Dosage Forms

- Capsule 1.5 mg, 3 mg, 4.5 mg, 6 mg

How to Dose

- Schizophrenia: initial 1.5 mg once daily; can increase to 3 mg once daily on day 2; should increase in 1.5–3 mg increments to reach therapeutic dose (recommended dose 1.5–6 mg once daily)
- Bipolar mania: initial 1.5 mg once daily; can increase to 3 mg once daily on day 2; can increase in 1.5–3 mg increments to reach therapeutic dose (recommended dose 3–6 mg once daily)
- Bipolar depression: initial 1.5 mg once daily; can increase to 3 mg once daily on day 15 if clinically warranted (recommended dose 1.5–3 mg once daily)



Dosing Tips

- Because of its long half-life, and the especially long half-life of one of its active metabolites, monitor for adverse effects and response for several weeks after starting cariprazine and with each dosage change; also washout of active drug will take several weeks
- Because of its long half-life, missing a few doses may not be as detrimental compared to other antipsychotics

- Dosing for bipolar depression is generally lower than for schizophrenia or bipolar mania
- For bipolar depression without mixed features, 1.5 mg/day dose may be adequate for many patients
- For bipolar depression with mixed features, 3.0 mg/day may be a better dose for some patients
- In general, the more manic features, the higher the dose in bipolar disorder, and dose can be adjusted according to this principle as the patient waxes and wanes symptoms of mania and depression
- Some patients with schizophrenia or acute bipolar mania may benefit from doses higher than 6.0 mg/day, as cariprazine has been studied up to 12 mg/day, but it is not approved for this use and side effects are definitely dose-related and in general therapeutic effects for the majority of patients do not justify dosing over 6.0 mg/day
- Slow titration from 1.5 mg/day to 3.0 mg/day in bipolar depression produced much less akathisia compared to rapid dose increases in studies of mania and schizophrenia
- Can be taken with or without food
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing

anticonvulsant, such as valproate, topiramate, or lamotrigine

- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Limited experience

Long-Term Use

- Approved for maintenance treatment of schizophrenia in adults
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Because clinical experience is lacking, down-titration may be prudent, especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- However, the long half-lives of cariprazine and its two active metabolites suggest that it may be possible to stop cariprazine abruptly

- See Switching section of individual agents for how to stop cariprazine
- Rapid discontinuation could theoretically lead to rebound psychosis and worsening of symptoms, but less likely with cariprazine due to its long half-life

Pharmacokinetics

- Metabolized by CYP450 3A4 into two long-lasting active metabolites
- Based on time to reach steady state, half-life for cariprazine is 2–4 days and for one of its active metabolites, didesmethyl cariprazine (DDCAR), is 1–3 weeks



Drug Interactions

- Initiating a strong CYP450 3A4 inhibitor in patients on a stable dose of cariprazine: reduce the current dose of cariprazine by half (for patients taking 4.5 mg, reduce to either 1.5 mg or 3 mg once daily; for patients taking 1.5 mg once daily, reduce to 1.5 mg every other day)
- Initiating cariprazine in patients taking a strong CYP450 3A4 inhibitor: administer 1.5 mg on day 1; do not dose on day 2; administer 1.5 mg on day 3 and on day 4; maximum dose 3 mg once daily
- Concomitant use of cariprazine and a CYP450 3A4 inducer is not recommended

- May increase effects of antihypertensive agents
- May antagonize levodopa, dopamine agonists

Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Monitor patients for activation of suicidal ideation, especially children and adolescents
- Dysphagia has been associated with antipsychotic use, and cariprazine should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy